

Constrained Risk Prediction

Hjalte Sørberg Mikkelsen

Steno Diabetes Center Copenhagen

Introduction

Accurate prediction of disease risk in the presence of competing risks remains a central challenge in clinical epidemiology. We target individualized predictions of the 5-year risk of type 2 diabetes (T2D) and simultaneously the 5-year risk of the competing risk of death. Common estimation methods, such as Fine-Gray regression and cause-specific Cox regression, target only one cause at a time and are not optimized for a given prediction time horizon. We develop a loss-based estimation framework for the simultaneous prediction of multiple competing risks at a prespecified time horizon. Our approach provides a way to estimate a broad class of existing regression models for competing risks such that the resulting models are optimized for individualized risk prediction. Crucially, the framework enables seamless integration of flexible learner libraries and enables structural constraints during estimation. These may include coherence constraints that guarantee that the predicted risks respect the parameter space and domain-informed shape constraints (e.g., monotonicity in age). Estimation is achieved via constrained optimization of a differentiable loss function, such as the Brier score.

The Problem

Assume we want to predict the risk of diabetes and the risk of death at some timepoint τ , i.e. we want to predict the risk of each cause in a competing risks scenario. Denote T as the time between time0 and the date of an event and $D \in \{0, 1, \dots, K\}$ as the cause of the event, where $D = 0$ is censoring. Let $X = (X^1, \dots, X^p)$ be a p -dimensional vector of baseline covariates, which can both be numeric or categorical.

Consider a setting where $D \in \{0, 1, 2\}$ and X is 1-dimensional. Define the cumulative hazard for $D = j$ as

$$\Lambda_j(t|x) = \exp(x\beta_j)\Lambda_{0j}(t)$$

where Λ_{0j} is the baseline cumulative hazard for cause j , which is defined as $\Lambda_{0j}(t) = \int_0^t \lambda_{0j}(s)ds$, where λ_{0j} is the baseline hazard. The baseline hazard is however not estimated when fitting a cause-specific cox model. Now define the survival function using the product integral as

$$S(t|x) = \prod_{s \leq t} (1 - d\Lambda_1(s|x) - d\Lambda_2(s|x)),$$

or using the exponential approximation

$$S(t|x) = \exp(-\Lambda_1(t|x) - \Lambda_2(t|x)).$$

We can then write the cumulative incidence function for cause j , $F_j(t|x) = P(T \leq t, D = j|X)$, as

$$F_j(t|x) = \int_0^t S(s-|x)d\Lambda_j(s|x).$$

The problem I will look at is that current methods of estimation of the cause-specific risks estimates the cause-specific cumulative incidence functions separately. The problem with this is that it can lead to wrong estimations, which can easily be verified as it can happen that for τ and some x^*

$$F_1(\tau|x^*) + F_2(\tau|x^*) > 1. \quad (1)$$

I will in this project develop a method to estimate the risks simultaneously such that the above can't happen. MAYBE ALSO MENTION THE PROPORTIONAL HAZARDS ASSUMPTION, AND THAT BY USING BRIER-SCORE AS ESTIMATOR WE DON'T NEED THAT ASSUMPTION AS WE OPTIMIZE THE PARAMETERS FOR A GIVEN τ , AND DONT OPTIMIZE FOR THE WHOLE TIME PERIOD. Maybe this should also be the new goal instead of the "estimation of all causes simultaneously" goal. i.e. "In this project we aim to develop an estimation method based on the proportional hazards model that do not need the proportional hazards assumption". This is interesting as the proportional hazards model is so popular and being able to still use it while having an assumption less is huge.

Different loss functions

Likelihood function

Asume we have N datapoints and K different causes. The general parametric likelihood function in a competing risks setting can be written as:

$$L = \prod_{i=1}^N \lambda_{j_i}(t_i|x_i)^{d_i} S(t_i|x_i),$$

where d_i is 1 if an event was observed and 0 if i is censored, and j_i is which event occurred for person i . Using that the survival function is a product of the individual cause-specific survival functions, $S(t_i|x_i) = \prod S_j(t_i|x_i)$, we can write the likelihood as:

$$\begin{aligned} L &= \prod_{i=1}^N \lambda_{j_i}(t_i|x_i)^{d_i} \prod_{j=1}^K \exp(-\Lambda_j(t_i|x_i)) \\ &= \prod_{i=1}^N \prod_{j=1}^K \lambda_{j_i}(t_i|x_i)^{d_{ij}} \exp(-\Lambda_j(t_i|x_i)). \end{aligned}$$

This means that the likelihood function is a product of K likelihoods, one for each cause. We can then maximize this likelihood by maximizing each cause-specific likelihood function, as long as the parameters are different for each cause.

Likelihood estimating equations

Assume two cause, only one covariate and constant baseline hazards. I then differentiate the log-likelihood functions with respect to the parameters. The negative log-likelihood function for cause j is then:

$$-\log(L_j) = \sum_{i=1}^N -d_{ij}(\beta_j x_i + \log(\lambda_{0j})) + t_i \exp(\beta_j x_i) \lambda_{0j}$$

and differentiated

$$\frac{d - \log(L_j)}{d\beta_j} = \sum_{i=1}^N -d_{ij} x_i + t_i x_i \exp(\beta_j x_i) \lambda_{0j}$$

$$\frac{d - \log(L_j)}{d\lambda_{0j}} = \sum_{i=1}^N -\frac{d_{ij}}{\lambda_{0j}} + t_i \exp(\beta_j x_i)$$

Brier score

As our target in risk prediction is to minimize the brier-score, it makes sense to fit the models by choosing the parameters that minimize the brier-score. Another good thing about the brier-score is that we use the cumulative incidence function directly in the score, which depends on all the cause specific cumulative hazard functions. This may solve the problem without needing a constraint (test).

Assuming we have N datapoints of the form $(\tilde{T}, \tilde{D}, X)$ where $\tilde{T}$ is the observed time (time to first observed event), $\tilde{D}$ is the observed event and X are the covariate vector. The Brier-Score for predicting cause j at timepoint τ can be written as

$$BS = \frac{1}{N} \sum_{i=1}^N (F_j(\tau|x_i) - o_i(\tau, j))^2,$$

where $o_i(\tau, j)$ is the true outcome of person i , i.e. $o_i(\tau, j) = 1(\tilde{D}_i = j, \tilde{T}_i \leq \tau)$. Assume K different causes and assume no censoring (with censoring just add a 0 state such that we have $K + 1$ states and the setup will be similar to that of the joint survival super learner). We then want to find the parameters that minimize this score i.e.

$$\begin{aligned} BSS &= \arg \min_{\beta \in \mathbb{R}, \Lambda_0 \in \mathbb{R}^+} \sum_{j=1}^K \frac{1}{N} \sum_{i=1}^N (F_j(\tau|x_i, \beta, \Lambda_0) - o_i(\tau, j))^2 \\ &= \arg \min_{\beta \in \mathbb{R}, \Lambda_0 \in \mathbb{R}^+} \frac{1}{N} \sum_{j=1}^K \sum_{i=1}^N \left(\int_0^\tau S(s|x_i, \beta, \Lambda_0) d\Lambda_j(s|x_i, \beta_j, \Lambda_{0j}) - o_i(\tau, j) \right)^2, \end{aligned}$$

where β are the beta coefficients for all causes and Λ_0 are the baseline hazards, with j indicating the coefficients for the specific cause. BSS = Brier-score-sum. This is actually just the Brier-score as it was written by Brier in the original paper.

Find estimating equations in Parametric Cox with exponential baseline hazard

Assume we are in a competing risks setting with two causes and one covariate x , and we are interested in predicting the risk of experiencing the events before or at timepoint τ using a cox model and let $\phi = \{\beta_1, \beta_2, \Lambda_{01}, \Lambda_{02}\}$. First we take the simpler example and use the exponential approximation of the survival function and assume a constant baseline hazard. The hazard for cause j will be

$$\lambda_j = \exp(\beta_j x) \lambda_{0j},$$

where λ_{0j} is some constant. We are interested in solving this:

$$BSS = \arg \min_{\beta_1, \beta_2 \in \mathbb{R}, \lambda_{01}, \lambda_{02} \in \mathbb{R}^+} \frac{1}{N} \sum_{j=1}^2 \sum_{i=1}^N (g_j(\phi) - o_i(\tau, j))^2.$$

First I rewrite $g_1(\phi) = \int_0^\tau S(s|x, \beta, \Lambda_0) \lambda_1(s|x, \beta_1, \Lambda_{01}) ds$ from the Brier-score written above (note that g_1 is equal to F_1 for a fixed timepoint τ):

$$\begin{aligned} g_1(\phi) &= \int_0^\tau \exp[-\exp(\beta_1 x) \Lambda_{01}(s-) - \exp(\beta_2 x) \Lambda_{02}(s-)] \exp(\beta_1 x) \lambda_{01} ds \\ &= \exp(\beta_1 x) \lambda_{01} \int_0^\tau \exp[-\exp(\beta_1 x) s^- \lambda_{01} - \exp(\beta_2 x) s^- \lambda_{02}] ds \\ &= \exp(\beta_1 x) \lambda_{01} \frac{\exp[\tau^- (-\exp(\beta_1 x) \lambda_{01} - \exp(\beta_2 x) \lambda_{02})] - 1}{-\exp(\beta_1 x) \lambda_{01} - \exp(\beta_2 x) \lambda_{02}} \\ &= \exp(\beta_1 x) \lambda_{01} \frac{1 - \exp[\tau^- (-\exp(\beta_1 x) \lambda_{01} - \exp(\beta_2 x) \lambda_{02})]}{\exp(\beta_1 x) \lambda_{01} + \exp(\beta_2 x) \lambda_{02}} \end{aligned}$$

As we are in continuous time $\tau^- = \tau$. Then we differentiate g_1 with respect to $\beta_1, \beta_2, \lambda_{01}$ and λ_{02} (to get the result for g_2 just swap all the 1's and 2's, also the ones on the left side of the equals sign):

$$\begin{aligned} \frac{\partial g_1(\phi)}{\partial \beta_1} &= \frac{x \lambda_{01} e^{x\beta_1}}{(\lambda_{01} e^{x\beta_1} + \lambda_{02} e^{x\beta_2})^2} \\ &\quad \cdot \left[\lambda_{02} e^{x\beta_2} \left(1 - e^{-\tau(\lambda_{01} e^{x\beta_1} + \lambda_{02} e^{x\beta_2})} \right) + \tau \lambda_{01} e^{x\beta_1} (\lambda_{01} e^{x\beta_1} + \lambda_{02} e^{x\beta_2}) e^{-\tau(\lambda_{01} e^{x\beta_1} + \lambda_{02} e^{x\beta_2})} \right] \\ \frac{\partial g_1(\phi)}{\partial \beta_2} &= \frac{x e^{\beta_1 x} \lambda_{01} e^{\beta_2 x} \lambda_{02}}{(e^{\beta_1 x} \lambda_{01} + e^{\beta_2 x} \lambda_{02})^2} \left[(1 + \tau(e^{\beta_1 x} \lambda_{01} + e^{\beta_2 x} \lambda_{02})) e^{-(e^{\beta_1 x} \lambda_{01} + e^{\beta_2 x} \lambda_{02})\tau} - 1 \right] \end{aligned}$$

$$\frac{\partial g_1(\phi)}{\partial \lambda_{01}} = \frac{e^{\beta_1 x}}{(e^{\beta_1 x} \lambda_{01} + e^{\beta_2 x} \lambda_{02})^2} \cdot \left[e^{\beta_2 x} \lambda_{02} \left(1 - e^{-\tau(e^{\beta_1 x} \lambda_{01} + e^{\beta_2 x} \lambda_{02})} \right) + \tau e^{\beta_1 x} \lambda_{01} (e^{\beta_1 x} \lambda_{01} + e^{\beta_2 x} \lambda_{02}) e^{-\tau(e^{\beta_1 x} \lambda_{01} + e^{\beta_2 x} \lambda_{02})} \right]$$

$$\frac{\partial g_1(\phi)}{\partial \lambda_{02}} = \frac{e^{\beta_1 x} \lambda_{01} e^{\beta_2 x}}{(e^{\beta_1 x} \lambda_{01} + e^{\beta_2 x} \lambda_{02})^2} \left[(1 + \tau(e^{\beta_1 x} \lambda_{01} + e^{\beta_2 x} \lambda_{02})) e^{-(e^{\beta_1 x} \lambda_{01} + e^{\beta_2 x} \lambda_{02})\tau} - 1 \right]$$

When implementing this in R all the things that can be inserted into $1 \exp()$ should be done so terms like $\exp^* \exp$ should be made into $1 \exp$ as otherwise they can explode or be $0 * \inf$ in R. Using this we can now differentiate the Brier-Score (BS) with respect to the betas to get the estimating equations:

$$\begin{aligned} \frac{\partial BSS(\phi)}{\partial \beta_1} &= \frac{1}{N} \sum_{j=1}^K \sum_{i=1}^N \frac{dg_j(\phi)}{\beta_1} 2(g_j(\phi) - o_i(t, j)) \\ &= 0 \end{aligned}$$

$$\begin{aligned} \frac{\partial BSS(\phi)}{\partial \beta_2} &= \frac{1}{N} \sum_{j=1}^K \sum_{i=1}^N \frac{dg_j(\phi)}{\beta_2} 2(g_j(\phi) - o_i(t, j)) \\ &= 0 \end{aligned}$$

$$\begin{aligned} \frac{\partial BSS(\phi)}{\partial \lambda_{01}} &= \frac{1}{N} \sum_{j=1}^K \sum_{i=1}^N \frac{dg_j(\phi)}{\lambda_{01}} 2(g_j(\phi) - o_i(t, j)) \\ &= 0 \end{aligned}$$

$$\begin{aligned} \frac{\partial BSS(\phi)}{\partial \lambda_{02}} &= \frac{1}{N} \sum_{j=1}^K \sum_{i=1}^N \frac{dg_j(\phi)}{\lambda_{02}} 2(g_j(\phi) - o_i(t, j)) \\ &= 0 \end{aligned}$$

Note that the $\frac{1}{N}$ terms can be removed.

Using Cox-Weibull

The hazard function for the Cox-Weibull model for cause j is

$$_j(t|x) = \lambda_j \exp(\beta_j x) v_j t^{v_j - 1},$$

where $\lambda_j > 0$ is the scale parameter and $v_j > 0$ is the shape parameter. The baseline hazard is

$$h_{0,j}(t) = \lambda_j v_j t^{v_j - 1},$$

and the cumulative baseline hazard function is

$$H_{0,j}(t) = \lambda_j t^{v_j}.$$

This means the survival function will be

$$S(t|x) = \exp(-(\exp(\beta_1 x) \lambda_1 t^{v_1} + \exp(\beta_2 x) \lambda_2 t^{v_2})),$$

and the cumulative incidence function for cause 1 is

$$F_1(t|x) = \int_0^t \exp(-(\exp(\beta_1 x) \lambda_1 s^{v_1} + \exp(\beta_2 x) \lambda_2 s^{v_2})) \lambda_1 \exp(\beta_1 x) v_1 s^{v_1-1} ds.$$

The cumulative incidence function does not have a closed form, so it needs to be numerical approximated in practise.

Estimation with constraint

Assuming we have N datapoints, the Brier-Score for predicting cause j at timepoint τ can be written as

$$BS_j = \frac{1}{N} \sum_{i=1}^N (F_j(\tau|x_i) - o_i(\tau, j))^2,$$

where $o_i(\tau, j)$ is the true outcome of person i (1 if event j is the observed event and has happened at or before timepoint t and 0 otherwise). Furthermore assume only two competing risks (T2D and death). We then want to find the parameters that minimize this score i.e.

$$\arg \min_{\beta \in \mathbb{R}} \sum_{j=0}^K \frac{1}{N} \sum_{i=1}^N (F_j(\tau|x_i, \beta) - o_i(\tau, j))^2.$$

In our setting $K = 2$ where $j = 0$ is the censored state and 1 and 2 are T2D and death. We need to minimize this score under some constraint, for example:

$$\sum_{j=0}^K F_j(\tau|x, \beta) \leq 1 \text{ for all } x,$$

or

$$\frac{d}{d\beta_{age}} F_j(\tau|x_i, \beta) > 0$$

for some j . In the T2D and death example $j \in \{1, 2\}$, as those risks should increase with age. In practise it may be easier to rewrite the minimization problem to:

$$\arg \min_{\beta \in \mathbb{R}} \sum_{j=0}^K \frac{1}{N} \sum_{i=1}^N (F_j(\tau|x_i, \beta) - o_i(\tau, j))^2 + I\left(\frac{d}{d\beta_{age}} F_j(\tau|x_i, \beta) \leq 0\right),$$

i.e. add the constraint to the loss function as a new term which will be 0 if the constraint holds and 1 otherwise which is large as we are using the Brier-score. We should probably multiply the indicator function with $K+1$, as we sum $K+1$ Brier-scores.

If we want methods like Newton-Raphson to work we probably need to use a different method as above, as the loss-function need to be smooth. Say we have the constraint $g(\beta) \leq 0$ then we can use the loss function:

$$\arg \min_{\beta \in \mathbb{R}} \sum_{j=0}^K \frac{1}{N} \sum_{i=1}^N (F_j(\tau|x_i, \beta) - o_i(\tau, j))^2 + \lambda \max(0, g(\beta))$$

where λ is some penalization parameter.

Non-convexity of BSS

The BSS can be shown to be non-convex...

A function f is convex if for any two points x and y in its domain and $\lambda \in [0, 1]$ the following holds:

$$f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y).$$

A convex functions second derivative must be non-negative across the entire domain, so it can be shown that a non-convex function will have atleast 1 point where

$$f''(x) < 0.$$

$$\begin{aligned} f(\beta, \Lambda_0) &= \sum_{j=1}^K \frac{1}{N} \sum_{i=1}^N (F_j(\tau|x_i, \beta, \Lambda_0) - o_i(\tau, j))^2 \\ &= \frac{1}{N} \sum_{j=1}^K \sum_{i=1}^N \left(\int_0^\tau S(s - |x_i, \beta, \Lambda_0) d\Lambda_j(s|x_i, \beta_j, \Lambda_{0j}) - o_i(\tau, j) \right)^2, \end{aligned}$$

How to handle censoring

Implement how to handle censoring in the simulation setup to test if it works and then write about it here. Its just the setup from the joint survival superlearner. Remember to use the Brier-score-sum for all causes+censoring when fitting, but then to show performance only use the Brier-score-sum for the two causes.

Simulation study

We want to test our method in three different setups: first where the model is correctly specified, second where the model is not correctly specified but

still proportional hazards and one where the proportional hazards assumption doesn't hold. We test our estimation method against the maximum likelihood estimator.

Correctly specified model

See file "`~/PhD-Project/Project1_constrainedriskprediction/Brier_scoreestimation/doesBSfinddatageneratingparameters.R`" for example.

Brier-score estimation does indeed find the data-generating parameters, but can fail if the starting parameters for the newton-raphson algorithm are chosen poorly.

Choosing optimizer